

Kosh

Simulates the whole plan. Keeps the working.

The household

where they stand, what if, what to do next
- with the reason behind every number

The adviser

which of 214 households needs a call this
week, ranked, with the evidence

The reviewer

what was advised, what else was
considered, why it lost - two years later

THE PROBLEM

Advice is easy to give. It is hard to evidence.

- Banking and investing apps show **what you have**, not **whether it's enough**.
- "What if I lose my job?" or "can we afford the car?" needs a spreadsheet most people never build.
- Advice is either expensive, or free and quietly selling a product.
- Generic AI chatbots will happily invent a number.

"We earn well. I just don't know if we're doing okay."

- the sentence every one of our test households opened with

Three households, three different problems

Ananya, 27 · Pune

₹50k/month unplanned surplus, food delivery up 35%, a 42% credit card.

Retirement odds 4% → 69% with one step-up setting.

Farhan & Zoya, 36 · Bengaluru

Home loan, one child, 3 months of emergency money, ₹50 L cover vs ₹3.8 Cr needed.

College fund 36% → 72%, retirement 16% → 92%.

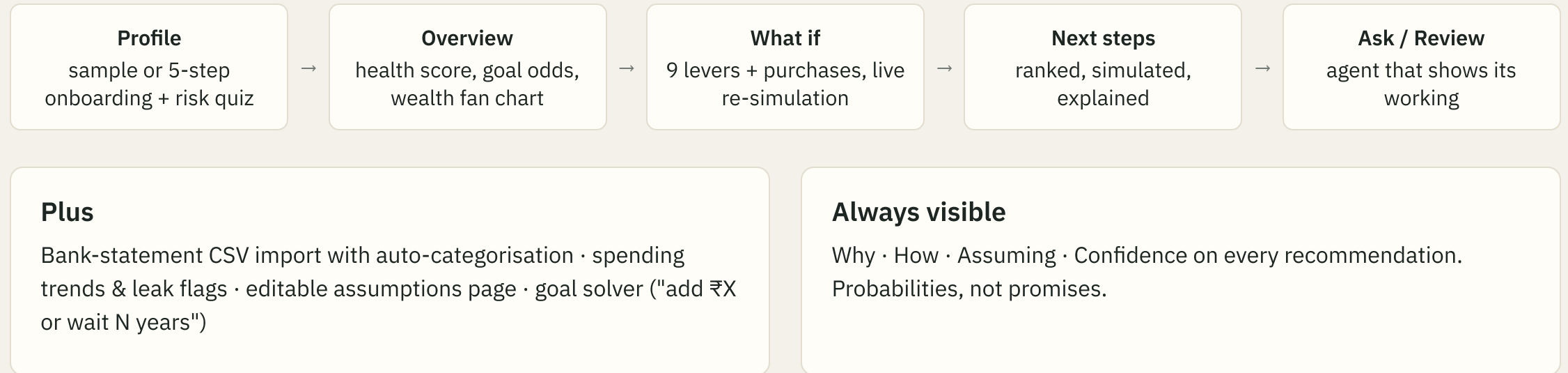
Mr. Iyer, 52 · Chennai

Saves 60% of income, nearly all in FDs and EPF. Six years to retirement.

Retirement 38% → 98% by automating his idle surplus.

All synthetic. Numbers from the live engine with default assumptions.

Five minutes from "am I okay?" to a plan



Every suggestion is simulated before it's shown

1. 11 generators spot situations: costly debt, thin cushion, idle cash, unplanned surplus, protection gaps, FD-heavy long-term money...
2. Each candidate is applied to a **copy** of the plan.
3. The whole plan is re-run - goals, health, cash flow.
4. Ranked by priority-weighted change in goal odds + safety + urgency.

Sheikh household · #1 "10% yearly step-up"

Retire at 60



Aarav's college fund



Same simulated markets before and after (common random numbers) - the difference is the change, not noise.

An adviser does not have one household

214 households. The same engine over every one — **171,200 market paths**, precomputed at build, served in 39ms.

DECISION	WHY
Base rates are not alerts	92% of this book is underfunded for retirement. Raising that 196 times buries the 11 with something fixable this week.
The queue has a capacity	An adviser can have ~12 real conversations a week. So the system picks 12 — and publishes the cutoff.
Every flag carries its numbers	"Protection gap" is an opinion. "1 dependent, no term cover, ₹2.03 Cr of need" is a meeting.

This week — top of the queue

- 1 · Gaurav & Radhika Shetty · ₹24.5 L
- Protection gap — no term cover, ₹2.03 Cr need
- Near-term goal — "Meera's education" at 0%, due 2030
- Off mandate — aggressive profile, 18% equity

Alert fatigue is how these systems die

A queue where everything is red is worse than no queue at all.

The alternatives were always there. We stopped throwing them away.

Ranking a recommendation **means** simulating every alternative against that household. At the moment of ranking, the engine is holding exactly what a reviewer will ask for in two years.

So the suitability record is a by-product: what was advised, what else was simulated, what each was projected to do, why it lost, every assumption relied on — and a SHA-256 digest, so alteration is detectable.

It generates evidence. It does not certify compliance, and it says so.

SEBI (IA) Regulations 2013, reg. 17

Reasonable basis for advice, and a record of that basis.

SEC Reg BI — Care Obligation

Reasonably available alternatives considered.

Why that clause is expensive

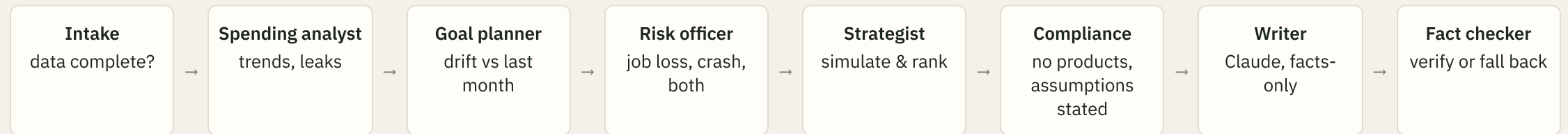
Firms struggle to evidence it because nobody writes the alternatives down. Here it falls out of how ranking works.

Claude plans and explains. The engine does the maths.

- **7 tools:** snapshot, run_scenario, check_goal, next_best_actions, analyze_spending, search_guides, get_assumptions.
- Multi-step tool loop; every step shown to the user.
- **Grounding check:** every ₹ and % in the answer must match a tool result. Untraced → sent back to the model once to fix.
- Refusal or outage → offline planner answers with the same tools.

```
◇ Understood: can I afford a purchase
→ get_snapshot   Health 56 · surplus ₹44,667/mo
→ run_scenario {big_purchase: ₹10 L, 1 yr}
    College fund 39% → 2%
→ search_guides "car loan emi"
✎ "You can pay for it, but it comes out of the college fund..."
✓ all 9 figures traced to calculations
```

The monthly review: eight specialists, one note



Proactive, not reactive: the review finds the problem before the user asks. Deterministic where it must be right, generative only where it must be human.

Assumptions are a page, not a footnote

Visible

Inflation 6%, education 9%, equity 11.5% $\pm$ 18%, surplus leakage 50%... each with a one-line reason.

Editable

Change any value and every probability, chart and recommendation re-runs.

Honest odds

We plan for ~80% odds, not the average (which is a coin flip). Goals keep their real asset mix - FDs stay FDs.

Categories, never products. Not a SEBI-registered adviser, and says so once - not on every line.

One container, or serverless on AWS

LAYER	SERVICE
Web	React on S3 + CloudFront (OAC)
API	Lambda (Node 22, arm64) via Function URL
AI	Claude on Amazon Bedrock
State	DynamoDB, on-demand, 30-day TTL
Ops	CloudWatch, X-Ray, error alarm
IaC	AWS SAM
CI/CD	GitHub Actions → OIDC → sam deploy → s3 sync → smoke test

Why a Function URL, not API Gateway?

Agent turns with several tool calls can exceed API Gateway's 30 s limit.

Why BM25, not a vector DB?

15 hand-written notes. Right tool for the size; Knowledge Bases is a one-function swap.

Same code everywhere

Laptop, Docker, this demo and Lambda run the identical Express app. The live demo is one container; the table is the production target, linted in CI.

Built so the numbers can be trusted

27

tests: engine properties,
agent, grounding, HTTP API

<0.6s

full dashboard: goals +
projection + 9 simulated
actions

0

numbers from the model that
aren't traced to a calculation

100%

demo works offline - no keys,
no Wi-Fi dependency

- Property tests: investing more never lowers odds; a crash never raises them; solver's answer actually reaches the target.
- Seeded Monte Carlo with common random numbers → deterministic, comparable, testable.

Where this goes

- **Banks & brokers:** an explainable planning layer inside existing apps - more SIP starts, step-ups and insurance attach, without product-pushing.
- **Employers:** financial wellness benefit; payroll data makes onboarding one click.
- **Advisers:** a first-pass plan and stress test before the meeting.

Next

- Account Aggregator (RBI AA framework) instead of CSV upload
- Tax engine: old vs new regime on real salary structure
- Historical bootstrap returns (fatter tails)
- Cognito auth, household sharing, monthly review by email

THANK YOU

**Kosh doesn't tell people what to buy.
It shows them what their choices do.**

kosh-2w6y.onrender.com

Repo · architecture doc · deployment guide - all in the submission.